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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2025

SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCPES100 –

MOVEMENT EDUCATION

(2024 ADMISSION ONWARDS)

PART- A

1) a) Movement that involve large muscle groups for activities like running and jumping

2)a) Repeating a dance sequence from memory

3)d) Static movement

4.b). Spatial awareness and planning.

5.c). The ability to use different parts of the body together to perform a task smoothly

6.a). Hopping

7.c). Reflexes, coordination, balance.

8.a). Iodine

9.b). It promotes children's emotional and social development

10)d) Reduced creativity

PART - B

Q11.Define physical literacy in the context of movement education

Physical Literacy:

1. Fundamental movement skills: Ability to perform basic movements like running, jumping, throwing. 2. Motor competence: Confidence and proficiency in various physical activities. 3. Physical fitness: Development of cardiovascular endurance, strength, flexibility. 4. Knowledge and understanding: Awareness of movement concepts, safety, and health. 5. Application and creativity: Ability to apply movement skills in different contexts.

1. Improved physical health: Enhanced fitness and reduced disease risk. 2. Increased confidence: Development of self-assurance and self-esteem. 3. Better motor skills: Improved coordination, balance, and overall movement ability.

Physical literacy is essential for lifelong physical activity and overall well-being.

Q12.Give an example of how you could integrate movement to improve a child's ability to recall information

Ans: Integrating Movement to Improve Recall:

1. Action word associations: Associate vocabulary words with specific actions.

2. Math movement sequences: Create movement sequences that represent math facts (e.g., jumping jacks for addition).

3. Historical dance routines: Create dance routines that depict historical events.

4. Kinesthetic storytelling: Use gestures and movements to retell stories.

Q13) How does movement education enhance children's learning experiences?

1. Multisensory engagement: Combines physical activity with cognitive learning. 2. Improved retention: Movement enhances memory and recall. 3. Increased motivation: Engaging and interactive learning experiences. 4. Develops problem-solving skills: Encourages creativity and critical thinking. 5. Promotes physical and emotional well-being: Supports overall health and self-esteem.

Benefits:1. Holistic development: Integrates physical, cognitive, and emotional growth. 2. Enhanced creativity: Encourages exploration and self-expression. 3. Better academic performance: Supports learning across various subjects.

Movement education can make learning more engaging, effective, and enjoyable for children.

Q14) what is immunity?

Ans: 1. The body's ability to resist and fight off infections and diseases. Innate immunity: Natural, non-specific defense mechanisms. 2. Adaptive immunity 3. Specific defense mechanisms that develop over time.

Q15) describe the importance of crawling in the early stages of movement development

Ans: 1. Motor skill development: Enhances coordination, balance, and strength.

2. Brain-body connection: Develops neural pathways and integrates sensory information.

3. Spatial awareness: Improves understanding of body position and movement.

4. Foundation for future skills: Prepares for more complex movements like walking and running.

Benefits:1. Strengthens muscles: Develops upper body, core, and leg strength. 2.Improves hand-eye coordination: Enhances visual-motor skills. 3. Enhances cognitive development: Supports problem-solving and exploration.

Crawling is a crucial stage in movement development, laying the foundation for future physical and cognitive abilities.

Q16) How do mutations in the growth hormone receptor gene (GHR) affect growth?

Ans: Mutations in the growth hormone receptor gene (GHR) can significantly impact growth and development.

Effects on Growth-Growth hormone insensitivity: Mutations in the GHR gene can lead to partial or complete insensitivity to growth hormone, resulting in growth deficiency and short stature.

- The mutations can alter the structure and function of the growth hormone receptor, affecting its ability to interact with growth hormone and trigger downstream signaling pathways.

Main gene factors are: Genes provide basic structure of the body

3types of gene factors

a) body type (height, weight, motor abilities) b) muscle fiber type (red muscle fiber, white muscle fiber, and combination of this fibers)

Conditional circumstances

Q17) How does the cell growth and division influence growth and development?

Ans:1) building blocks: macro and micro nutrients help produce cells. During rapid growth nutrients play vital role.

2) tissue repair and maintenance: Bone, muscles, skin, organs etc. are repaired

3)Brain development and function: Brain structure maintenance, cognitive development

4) Immune cell production (production of white blood cells) Reduce risk of infections

5)Hormone production and regulations, Growth hormone production, Thyroid hormone production

Q18) how does inclusion benefit differently abled individuals in physical education?

Ans: 1) Improved physical health, 2) enhanced motor skills3), cognitive development, 4) social and emotional well-being, 5) independence, 6) stress reduction.

Q19) explain how problem solving is encouraged through sports and games?

Ans: 1. Players must adapt to changing situations., 2. Teams develop strategies to outmaneuver opponents, 3. Quick decision-making: Players make rapid decisions during gameplay.

For example, connecting with football or other game: Reading the game: Players anticipate opponents' movements.

Overcoming obstacles: Players find ways to bypass opponents' defenses.

Benefits: 1. Improved critical thinking: Players develop problem-solving skills. 2. Enhanced creativity: Players find innovative solutions. 3. Better teamwork: Collaboration and communication improve.

Sports like football encourage problem-solving, promoting cognitive development and teamwork.

Q20) What is the role of assistive technology in adapted physical activities?

Ans: 1. Enabling participation: Assistive technology helps individuals with disabilities participate in physical activities.

2. Enhancing accessibility: Adaptive equipment and technology make physical activities more accessible.

3. Promoting independence: Assistive technology empowers individuals to engage in physical activities with greater autonomy. Examples:1. Adaptive sports equipment: Modified wheelchairs, prosthetics. 2. Assistive devices: Walkers, canes., 3. Technology integration: Virtual reality, exergaming.

Benefits:1. Increased inclusion: More individuals can participate, 2. *Improved physical fitness*: Regular physical activity, 3. *Enhanced confidence*: Greater autonomy and independence.

Assistive technology plays a vital role in promoting inclusive and accessible physical activities.

Q21) how can motion capture technology be used to improve a child's motor skills?

1. Personalized feedback: Provides detailed analysis of movements. 2. Real-time tracking: Monitors and adjusts movements in real-time. 3. Engaging exercises: Interactive games and activities promote practice.

Benefits:1. Improved coordination: Enhances motor control and precision. 2. Enhanced learning: Accelerates skill acquisition. 3. Increased motivation: Interactive and fun learning experiences.

Applications: Sports training: Enhances athletic performance. Education: Integrates technology into physical education. Motion capture technology offers an innovative approach to improving motor skills.

Q22) How does VR make physical activity more engaging for children?

Immersive training: VR can create realistic experiences

Pain management: by not feeling pain we understand how it affects (aviation stimulator)

Motor skill learning and accessibility

PART- C

Q23) explain the concept of physical literacy and its importance in lifelong well-being

Ans: The ability to move with confidence, competence, and enjoyment in various physical activities.

1. Motor skills: Fundamental movement skills. 2. Movement confidence: Believing in one's ability. 3. Knowledge: Understanding physical activity benefits.

Importance:

1. Lifelong physical activity: Encourages sustained engagement.2. *Overall well-being: Supports physical, mental, and social health. 3. Improved health outcomes: Reduces chronic disease risk.

Benefits:

1. Increased physical activity: Promotes regular exercise. 2. Enhanced self-esteem: Builds confidence. 3. Better mental health: Reduces stress and anxiety.

Physical literacy is essential for lifelong well-being, enabling individuals to engage in physical activities with confidence and enjoyment.

Q. 24) Explain the role of attention in cognitive devolopment and how movement activities help improve sustained attention.

Ans:1. Focus and concentration: Attention enables learning and information processing. 2. Filtering distractions: Attention helps prioritize relevant information. 3. Regulation of cognitive resources: Attention allocates resources for tasks.

Movement Activities and Sustained Attention

1. Improved focus: Regular physical activity enhances attentional abilities. 2. Reduced distractions: Movement activities promote self-regulation, 3. Enhanced cognitive flexibility: Movement supports adaptability.

Benefits:1. Better academic performance: Improved attention supports learning. 2. Increased productivity: Enhanced focus and concentration. 3. Improved self-regulation: Better emotional control.

Movement activities play a significant role in developing sustained attention, benefiting cognitive development and overall well-being.

Q25) Explain the role of nutrition in the development stages of a child.

Ans: 1. Growth and development: Provides essential nutrients for physical growth.

2. Energy and strength: Supports daily activities and overall health.

3. Cognitive development: Nutrients support brain function and development.

Carbohydrates, protein and fats from macro nutrients and

Vitamins and minerals from micro nutrients (Vital for various bodily functions)

Importance:1. Optimal growth: Supports healthy development, 2. Improved cognitive function: Enhances learning and focus, 3. Strong immune system: Helps fight off infections.

Adequate nutrition plays a vital role in supporting the overall development and health of children.

Evolution of child movements during growth are: Reflexive movements(infancy), Fundamental movements(childhood), Specialized movements (pre-school), Refined movements (school age).

Q26) explain the significance of the Moro reflex in infant neurological development

Ans :1. Neurological assessment: The Moro reflex is an indicator of neurological integrity.

2. Brainstem function: The reflex is controlled by the brainstem, which regulates vital functions.

3. Developmental milestone: The presence and strength of the Moro reflex can indicate normal neurological development.

Importance:

1. Assessment tool: Healthcare professionals use the Moro reflex to assess neurological function.

2. Indicator of potential issues: Abnormalities in the Moro reflex may indicate neurological problems. The Moro reflex plays a significant role in assessing infant neurological development and function.

Q27) Discuss the role of sensory processing in the development of gross motor skills during early childhood.

Ans: 1. Integration of sensory inputs: Sensory processing enables children to integrate sensory information from their environment, including proprioception (body position and movement), vestibular input (balance and spatial awareness), and tactile input (touch and texture).

2. Motor planning and execution: Sensory processing supports the development of motor plans, allowing children to execute movements effectively.

3. Adaptation and refinement: Children refine their movements based on sensory feedback, adjusting their actions to achieve desired outcomes.

Importance of Sensory Processing:

1. Foundational for motor skills: Sensory processing lays the groundwork for gross motor skills.

2. Enhances coordination and balance: Accurate sensory processing supports coordinated movements.

3. Supports learning and development: Sensory integration facilitates exploration and learning.

Impact on Gross Motor Skills:

1. Walking and balance: Sensory processing helps children develop stability and balance.

2. Running and jumping: Accurate sensory feedback enables children to adjust their movements.

3. Climbing and navigating: Sensory processing supports spatial awareness and motor planning.

Sensory processing plays a vital role in the development of gross motor skills during early childhood.

Q28. How does Al contribute to adaptive learning in child movement education?

Ans:

1. Personalized feedback: AI provides tailored feedback. 2. Real-time analysis: AI analyzes movement patterns. 3. Customized exercise: AI generates adaptive exercises.

Benefits:1. Improved motor skills: Enhanced gross motor development. 2. Increased engagement: Interactive and fun learning experiences. 3. Efficient learning: Adaptive learning optimizes progress

Applications:

1. Movement tracking: AI tracks and analyzes movements. 2. Game-based learning: AI-powered games promote engagement. 3. Virtual coaching: AI provides guidance and support.

AI enhances child movement education through adaptive learning

Injury prevention, Performance Enhancement, Movement analysis, Rehabilitation, Biofeedback